

Ponjeyakumar R

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Integrated M.Tech Software Engineering · Frontend & AI Systems · Full-Stack Development

SUMMARY

Final-year Integrated M.Tech student at VIT Chennai (9.06 CGPA) specializing in React development and full-stack engineering. Experienced in building responsive web applications, interactive dashboards, and production-ready sites. Strong background in API integration, real-time data handling, and scalable system design.

EDUCATION

Integrated M.Tech in Software Engineering

Vellore Institute of Technology

- Cumulative GPA: **9.06 / 10.0**

2022 – 2027

Chennai, TN

Class XII: 92.5% — **Class X: 94.6%**

Xan Matric Higher Secondary School

2020 – 2022

Tuticorin, TN

TECHNICAL SKILLS

Frontend

React 18, JavaScript (ES6+), TypeScript, Tailwind CSS, HTML5, CSS3

Backend & Tech

FastAPI, Spring Boot, Node.js, REST APIs, WebSockets, JWT

AI/ML & Data

PyTorch, FAISS, scikit-learn, Pandas, NumPy, RAG Pipelines

Infrastructure

AWS, Docker, Git, MySQL, MongoDB, Redis, Vercel

Tools

Postman, Agile, Scrum, Figma

EXPERIENCE

Research Intern

Centre for Cyber Physical Systems (CCPS), VIT Chennai

Apr 2025 – Sep 2025

Chennai, TN

- Developed a deforestation detection framework utilizing high-resolution satellite imagery.
- Built an automated data pipeline that improved change-detection accuracy by **20%**.
- Integrated multi-source optical data to reduce false-positive rates by **15%**.

PROJECTS

AI Knowledge Assistant (RAG Pipeline)

2025

React, FastAPI, FAISS, PyTorch

- Built a semantic search system using vector embeddings for sub-second document retrieval.
- Implemented a streaming chat interface with markdown rendering for AI responses.

Secure URL Shortener

2024

Spring Boot, MySQL, Redis, JWT

- Designed a high-performance shortening service with Redis caching to minimize DB latency.
- Secured endpoints with JWT authentication and rate-limiting for DDoS protection.

Collaborative Document Editor

2025

Java, Spring Boot, WebSocket, Redis Pub/Sub

- Developed a real-time collaborative editor using WebSocket for live multi-user synchronization and conflict resolution.
- Implemented Redis Pub/Sub for event broadcasting across sessions; added version control to track document history.

Cloud-Based Attendance System

2025

AWS, MySQL

- Deployed a role-based access application on AWS with automated serverless notifications via Lambda and SNS.
- Optimized MySQL schema design and query structure, reducing execution time by **30%**.

PUBLICATIONS & CERTIFICATIONS

- **Pub:** Deforestation Detection Using Multi-Source Satellite Data
- **Pub:** Algal Bloom Detection Using Satellite Imagery
- **Certifications:** Agile Scrum — AWS Cloud Security — CCPS Internship

Under Review
Published, 2025